Demand Forecasting for Retail Stores

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree of

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PRINCIPLES OF DEEP NEURAL COMPUTATION (24GMI3101HF)

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Methodology

The proposed demand forecasting system is designed to predict future product demand in retail stores using historical sales data and modern data analysis techniques. Accurate demand forecasting helps retailers maintain proper stock levels, reduce wastage, and improve customer satisfaction. The methodology combines **time series analysis with deep learning techniques** to create a reliable forecasting system. This hybrid approach improves prediction accuracy and allows the system to adapt to different retail environments.

The methodology consists of several stages including **data collection, data preprocessing, feature engineering, model development, training, prediction, evaluation, and system accessibility**. Each stage plays an important role in building an efficient demand forecasting model.

1. Data Collection

The first step in the methodology is collecting relevant retail sales data. Historical data is very important for predicting future demand because it helps identify patterns and trends in product sales.

The dataset used in this project contains several types of information such as:

- Product ID or product name
- Sales quantity
- Date of sale
- Promotional offers
- Holiday information
- Seasonal sales trends

Retail stores usually generate large amounts of data every day through billing systems and inventory management systems. This historical data is used as the main input for the forecasting model.

The collected dataset provides information about how customer demand changes over time. For example, some products may have higher demand during weekends, festivals, or holiday seasons. By studying these patterns, the forecasting model can learn how demand behaves in different situations.

Proper data collection is important because the accuracy of the forecasting system depends heavily on the quality and quantity of the data used.

2. Data Preprocessing

The raw data collected from retail systems may contain errors, missing values, duplicate entries, or inconsistent formats. If such data is directly used for training the model, it may reduce prediction accuracy.

Therefore, **data preprocessing** is performed to clean and prepare the dataset before applying forecasting techniques.

The main preprocessing steps include:

Removing Duplicate Records

Sometimes the same sales record may appear multiple times in the dataset. Duplicate entries are removed to avoid incorrect analysis.

Handling Missing Values

In some cases, certain fields such as sales quantity or dates may be missing. These missing values are handled by replacing them with appropriate values such as averages or previous data.

Converting Dates into Time Series Format

The date column is converted into a time-series format so that the model can analyze sales patterns over time. Additional information such as day of week, month, or season can also be extracted from the date.

Data Normalization

Numerical values such as sales quantity are normalized or scaled to a standard range. This helps the deep learning model process the data more efficiently.

Data preprocessing improves the quality of the dataset and helps the forecasting model learn patterns more effectively.

3. Feature Engineerin

Feature engineering is an important step in demand forecasting because it helps the model understand the factors that influence product demand.

A **feature** is a variable or attribute that affects sales patterns. In retail forecasting, many factors can influence demand. Therefore, selecting the right features helps improve prediction accuracy.

Some of the important features used in this project include:

- Historical sales data
- Seasonal trends
- Promotional activities
- Holiday effects
- Product category information

For example, demand for certain products may increase during festivals or promotional discounts. By including such factors in the dataset, the forecasting model can learn how these events affect sales.

Feature engineering also helps in identifying relationships between different variables. This allows the model to make more accurate predictions.

4. Hybrid Forecasting Model Development

To improve forecasting accuracy, the proposed system uses a **hybrid forecasting approach**. This approach combines **time series analysis with deep learning techniques**.

Traditional forecasting methods rely only on statistical models. However, these models may not perform well when demand patterns become complex. Therefore, deep learning models are used to capture complex relationships in the data.

Time Series Analysis

Time series analysis is used to study sales data over time. It helps identify patterns such as:

- Long-term trends
- Seasonal variations
- Cyclic demand patterns

For example, some products may sell more during certain months or weekends. Time series analysis helps detect these repeating patterns.

Deep Learning Model – LSTM

The deep learning model used in this project is **Long Short-Term Memory (LSTM)**. LSTM is a special type of neural network designed for time-series data.

LSTM networks are capable of learning long-term dependencies in data. This means they can remember past sales patterns and use that information to predict future demand.

For example, if a product usually sells more during a specific festival season, the LSTM model can learn this pattern and forecast higher demand during the same period in the future.

By combining time series analysis with LSTM networks, the hybrid forecasting system achieves better performance than traditional forecasting methods.

5. Model Training

After developing the forecasting model, the next step is **training the model using historical sales data**.

The dataset is divided into two parts:

- **Training Dataset**
- **Testing Dataset**

The training dataset is used to teach the model how demand changes over time. During training, the model analyzes past sales patterns and learns relationships between different features.

The testing dataset is used to evaluate the model's prediction accuracy. It helps verify whether the model can correctly predict demand for unseen data.

During training, the model parameters are adjusted automatically to reduce forecasting errors. This process continues until the model achieves acceptable performance.

Model training is an important stage because it determines how well the system can learn patterns from historical data.

6. Demand Prediction

Once the model is trained successfully, it is used to predict future product demand.

The forecasting model analyzes recent sales patterns and generates predictions for upcoming time periods such as:

- Daily demand
- Weekly demand
- Monthly demand

These predictions help retailers plan their inventory levels more effectively.

For example, if the system predicts higher demand for a particular product next week, the retailer can order additional stock in advance. Similarly, if demand is expected to decrease, the retailer can avoid purchasing excess inventory.

Demand prediction helps retailers make better decisions related to supply chain management and inventory planning.

7. Performance Evaluation

After generating demand predictions, it is important to measure the performance of the forecasting model. This helps determine how accurate the predictions are.

In this project, model performance is evaluated using common forecasting evaluation metrics such as:

Mean Absolute Error (MAE)

MAE measures the average difference between predicted values and actual sales values. Lower MAE values indicate better forecasting accuracy.

Root Mean Square Error (RMSE)

RMSE measures the square root of the average squared differences between predicted and actual values. It gives more importance to larger prediction errors.

By calculating these metrics, the effectiveness of the forecasting model can be evaluated. Lower error values indicate that the model is producing accurate predictions.

Performance evaluation also helps identify areas where the model can be improved.

8. Dissemination and Accessibility

An important goal of this project is to make the forecasting system accessible and easy to use.

The demand forecasting model can be implemented using programming tools such as **Python** along with machine learning libraries like:

- TensorFlow
- Keras
- Scikit-learn
- Pandas and NumPy

The developed model and source code can be shared through **GitHub repositories**. This allows other researchers, developers, and students to access the system, study the methodology, and apply it to different retail applications.

Providing open access to the implementation improves collaboration and helps improve the forecasting model over time.

The system can also be adapted to different retail environments such as supermarkets, online stores, and supply chain management systems.

Summary of Methodology

The proposed methodology combines data analysis, time series techniques, and deep learning models to create an effective demand forecasting system. The hybrid approach improves forecasting accuracy and allows the system to adapt to various retail environments.

By using this methodology, retailers can better understand customer demand patterns, reduce inventory costs, and improve business performance.